



EDU TERIA

72nd BPSC

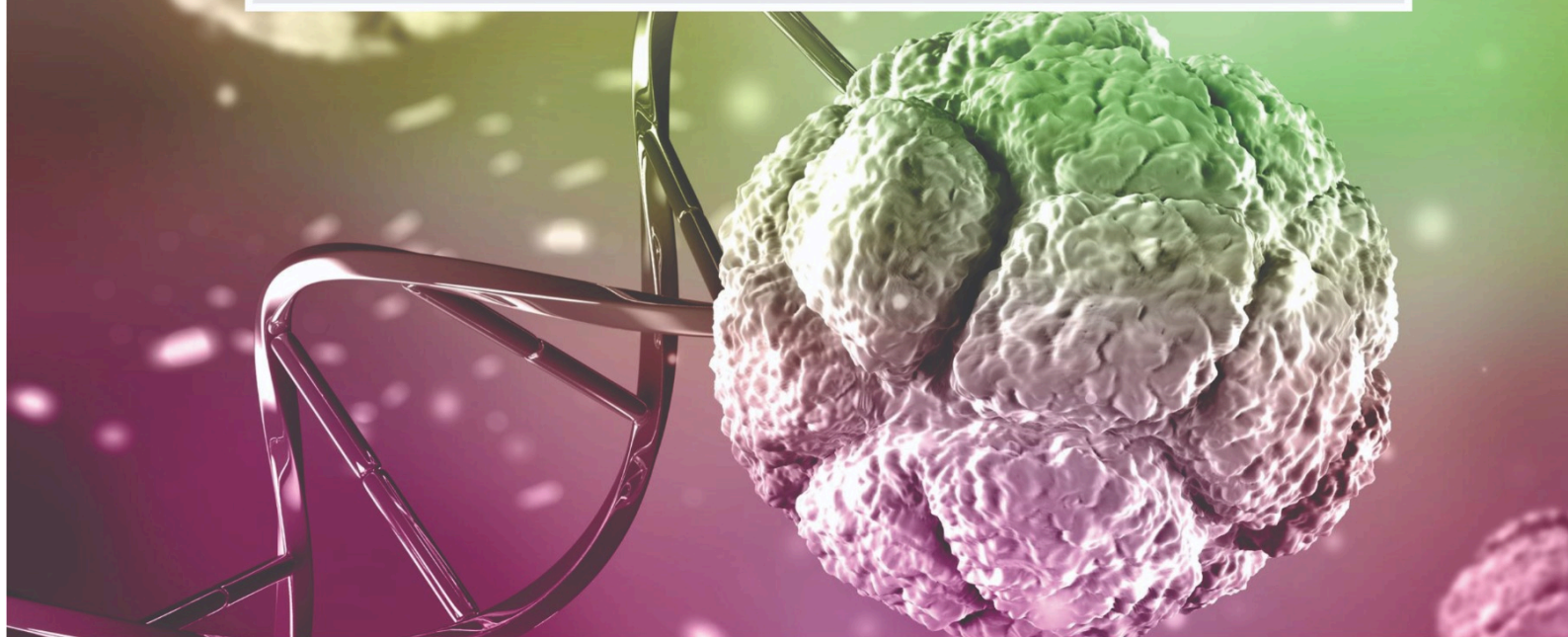
Prelims Online Batch

BIOLOGY

LECTURE

6

Nucleic Acid-DNA RNA and Protein Synthesis



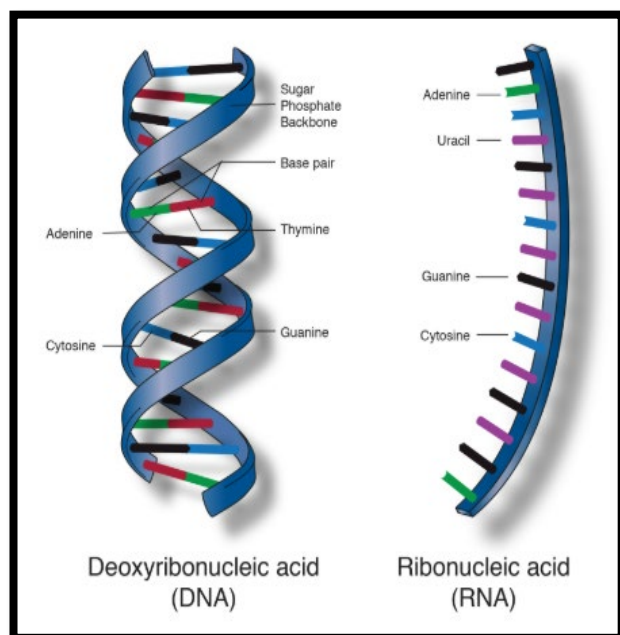
Nucleic Acid-DNA RNA and Protein Synthesis

Nucleic Acid

- F. Meischer discovered nucleic acid in nucleus of pus cell and called it "nuclein". The term nucleic acid was coined by "Altman."

- Nucleic acids are polymer of nucleotides.

Example : DNA and RNA.



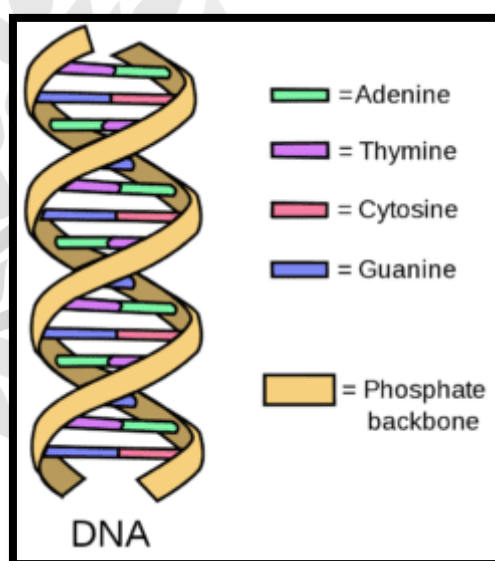
- Nucleotide = Nitrogen base + pentose sugar + phosphate
- Nucleoside = Nitrogen base + pentose sugar.
- A. Nitrogen base : On the basis of structure nitrogen bases are broadly of two types :-
 - Pyrimidines** - Consist of one pyrimidine ring. Skeleton of ring composed of two nitrogen and four Carbon atoms. e.g. Cytosine, Thymine and Uracil.
 - Purines** - Consist of two rings i.e. one pyrimidine ring ($2N + 4C$) and one imidazole ring ($2N + 3C$) e.g. Adenine and Guanine.
 - Pentose Sugar** (Number of carbon = 5)
 - Phosphate :**
 - Acidic
 - Negative charged
- Nitrogen base forms bond with first carbon of pentose sugar to form a nucleoside. Nitrogen of first place (N1) forms bond with sugar in case of

pyrimidines while in purines nitrogen of ninth place (N9) forms bond with sugar.

Genetic Material

DNA

DNA is the hereditary material present in almost all living organisms. It carries the genetic instructions required for the growth, development, functioning, and reproduction of organisms. DNA as an acidic substance present in nucleus was first identified by Friedrich Meischer in 1869.



- DNA term was given by - Zacharis
- DNA is long polymer of deoxyribonucleotides.
- DNA is negatively charged.

In DNA pentose sugar is deoxyribose sugar and four types of nitrogen bases **A, T, G, C**

With the help of this study, Watson and Crick (1953) proposed a double helix model for DNA. For this model Watson, Crick and Wilkins were awarded Noble Prize in 1962.

- One main hallmark (main point) of double helix model is complementary base pairing between purine and pyrimidine.
- According to this model, DNA is composed of two polynucleotide chains.
- Both polynucleotide chains are complementary and antiparallel to each other.



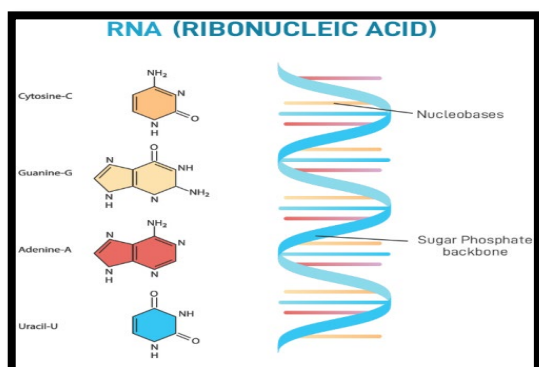
- In both strand of DNA direction of phosphodiester bond is opposite. i.e. If direction of phosphodiester bond in one strand is 3'-5' then it is 5'-3' in another strand.
- Both strand of DNA are held together by hydrogen bonds. These hydrogen bonds are present between nitrogen bases of both strand.
- Adenine binds to thymine by two hydrogen bonds and cytosine binds to guanine by three hydrogen bonds.
- In a DNA molecule one purine always pairs with a pyrimidine. This generates approximately uniform distance between the two strands of DNA.
- In DNA plane of one base pair stacks over the other in double helix. This, in addition to H-bonds, confers stability of the helical structure of DNA.

Functions of DNA

1. Genetic Information Storage DNA stores hereditary information that determines the traits of organisms.
2. Replication DNA can replicate itself during cell division so that genetic information is passed to daughter cells.
3. Protein Synthesis DNA controls protein synthesis through RNA. This concept is explained by the Central Dogma of Molecular Biology : **DNA → RNA → Protein**
4. Mutation and Evolution Changes in DNA sequence lead to mutations, which are responsible for genetic variation and evolution

RNA

RNA is a nucleic acid that plays a crucial role in protein synthesis and gene expression. Unlike DNA, RNA is usually single-stranded.



Structure of RNA is fundamentally same as DNA, but there are some differences. The differences are as follows :—

1. In place of De-oxyribose sugar in DNA, there is present Ribose sugar in RNA.
2. In place of nitrogen base Thymine in DNA, there is present uracil in RNA.
3. RNA is made up of only one polynucleotide chain i.e. R.N.A. is single stranded.

Exception :—

- RNA found in Reo - virus is double stranded, i.e. it has two polynucleotide chains.

Types of RNA :

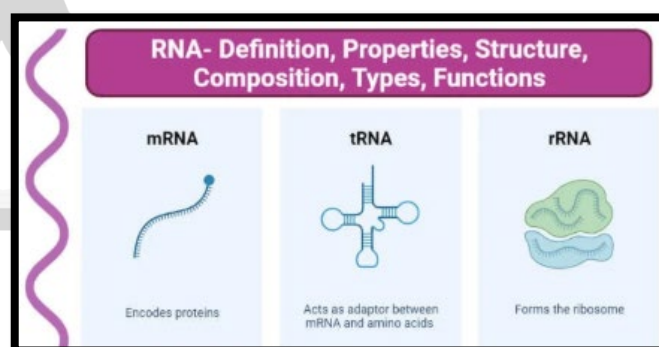
1. **Genetic RNA or Genomic RNA** - In some viruses RNA works as genetic material and it transfers information from one generation to next generation. eg. Reo virus, TMV, QB bacteriophage.
2. **Non-genetic RNA** - mainly of 3 types -
(1) **r** - RNA (2) **t** - RNA (3) **m** - RNA

1. Ribosomal RNA (r - RNA) :—

- This RNA is 80% of the cell's total RNA
- r RNA was discovered by Kuntze.
- It is found in ribosomes and it is produced in nucleolus.
- It is the most stable form of RNA

Function :—

- At the time of protein synthesis, r-RNA provides attachments site to t-RNA and m-RNA and attaches them on the ribosome.



2. Transfer - RNA (t-RNA) :—

- It is 10-15% of total RNA.
- It is synthesized in the nucleus by DNA.
- It is also known as soluble RNA (sRNA)
- It is also known as Adapter RNA.
- It is the smallest RNA (4s).

Function:– At the time of protein synthesis it acts as a carrier of amino-acids.

3. Messenger RNA (m –RNA) :-

- The m - RNA is 1 - 5% of the cell's total RNA.
- The name m-RNA was given by Jacob and Monod.
- The m - RNA is produced by genetic DNA in the nucleus.
- It is least stable RNA.

Difference Between DNA and RNA		
Feature	DNA (Deoxyribonucleic Acid)	RNA (Ribonucleic Acid)
Structure	Double-stranded helix	Usually single-stranded
Sugar	Deoxyribose	Ribose
Bases	A, T, G, C	A, U, G, C
Function	Long-term storage of genetic information	Protein synthesis and transmission of genetic code
Location	Nucleus, Mitochondria	Nucleus, Cytoplasm, Ribosomes
Stability	More stable	Less stable

Protein Synthesis

Protein synthesis is the biological process by which cells produce proteins using the genetic information stored in DNA. Proteins are essential for cell structure, enzymes, hormones, and body functions. The flow of genetic information during protein synthesis follows the

Central Dogma of Molecular Biology:

DNA → RNA → Protein

Protein synthesis mainly occurs in two major stages:

- Transcription
- Translation

1. **Transcription:** Transcription is the process in which genetic information from DNA is copied into messenger RNA
2. **Translation:** Translation is the process by which the genetic code in mRNA is converted into a sequence of amino acids, forming a protein.

Importance of Protein Synthesis

- Production of enzymes
- Formation of structural proteins
- Synthesis of hormones
- Cell growth and repair
- Regulation of metabolism

